Variability and acceleration in the growth of children's early vocabulary

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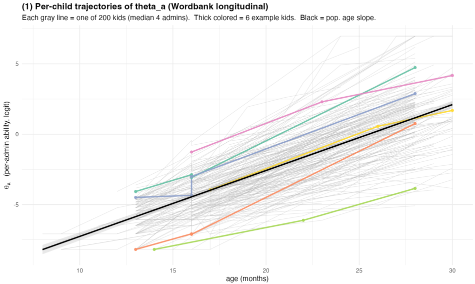
2026

Vocabulary growth: rapid, variable, consequential

Theoretical: Growth mechanisms are diagnostic of how children's learning works — and how it changes with development.

Practical: Variation in vocabulary growth is a key precursor of academic outcomes.

Methodological: Vocabulary count has a property nearly no other psychological metric has — a *true zero*. Children at birth know zero words. Absolute variability in word counts is interpretable, not just z-scores.



Per-admin theta on Wordbank longitudinal sample. Each grey line = one child.

Two signatures of vocabulary growth

What any theory of word learning has to predict

Variability

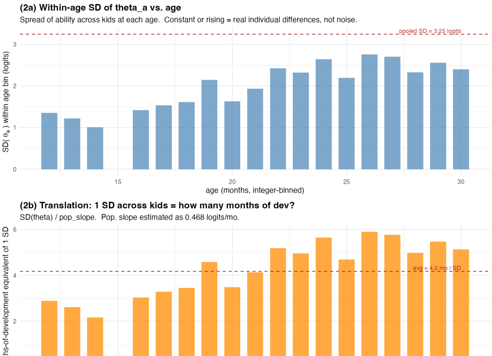
Children fan out, not converge.

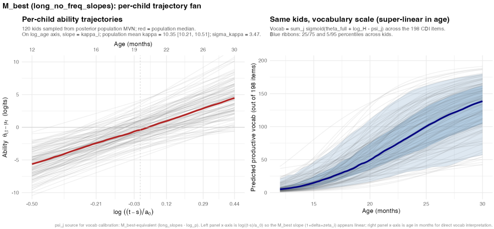
Within-age SD of ability *grows* with age.

Acceleration

Vocabulary grows super-linearly in tokens.

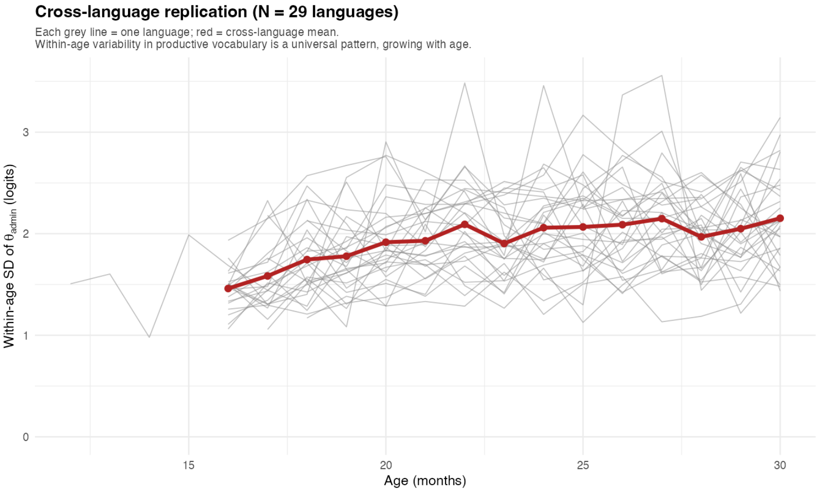
Population scaling exponent $\kappa_{\text{pop}} \gg 1$.





Cross-language replication

Variability and growth-with-age across Wordbank languages



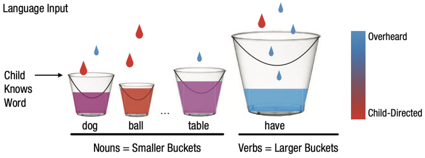
The structural null: pure accumulator

Pure accumulator. Each child accumulates word-evidence at per-child input rate r_i; word j is acquired when difficulty δ_j is crossed.

Scaling exponent κ_i: ability scales as $(\text{tokens})^{\kappa_i}$.

- $\kappa_i = 1$ for every child: pure unit accumulator.
- Between-child variance comes only from σ_r (input rate).
- No per-child slope variation.

Two estimands diagnose deviation: $\boldsymbol{\sigma_\alpha}$ (efficiency variation) and the $\boldsymbol{\kappa_i}$ posterior (*is κ above 1, and does it vary?*).



Bucket-fill metaphor: each child hears tokens at rate $r_i \cdot p_j$; word j is acquired when its bucket fills past difficulty δ_j (Kachergis et al. 2021).

Where prior accumulator models sit

None has both σ_α and σ_κ free

Model	per-child σ_α	per-child σ_κ	$\kappa_{\mathbf{pop}} > 1$ via
McMurray (2007) pure accumulator	–	–	only via $g'(\tau)$
Mitchell & McMurray (2009) + leverage	–	–	shifts but cannot create
Hidaka (2013) closed-form AoA	–	–	rate-change exponent D
Mollica & Piantadosi (2017) Gamma	–	–	embedded in per-word λ_w
Kachergis et al. (2021) 1-PL Rasch	θ_i free	–	linear age covariate
M_best (this work)	**free**	**free**	**free**

Common gap. No prior accumulator model has both σ_α *and* σ_κ free. Most have neither. The two signatures we measure — variability and acceleration — are precisely what these models cannot produce.

The model: a progressive build

Rasch IRT + accumulator dynamics, in three steps

Kid i, word j: $P(\text{produces}_{ij}) = \sigma(\eta_{ij})$, with $\eta_{ij} = \theta_i - \delta_j$ (ability − difficulty).

Step 1 – Pure accumulator. $\theta_i = \log r_i + \log t \iff \exp(\theta_i) = r_i \cdot t$
Ability = rate × time. Variation between kids comes only from r_i.

Step 2 – + per-child efficiency α_i. $\theta_i = \log r_i + \log \alpha_i + \log t \iff \exp(\theta_i) = \alpha_i \cdot r_i \cdot t$
Each child has a multiplicative efficiency over and above input rate.

Step 3 – + per-child scaling exponent κ_i. $\theta_i = \log r_i + \log \alpha_i + \kappa_i \log t \iff \exp(\theta_i) = \alpha_i \cdot r_i \cdot t^{\kappa_i}$
Children also differ in how steeply ability scales with cumulative input.

Two diagnostic questions: $\pi_\alpha = \dfrac{\sigma_\alpha^2}{\sigma_\alpha^2 + \sigma_r^2} > 0.5\,?$ (efficiency-share of variance) $\kappa_i > 1\,?$
(super-linear scaling)

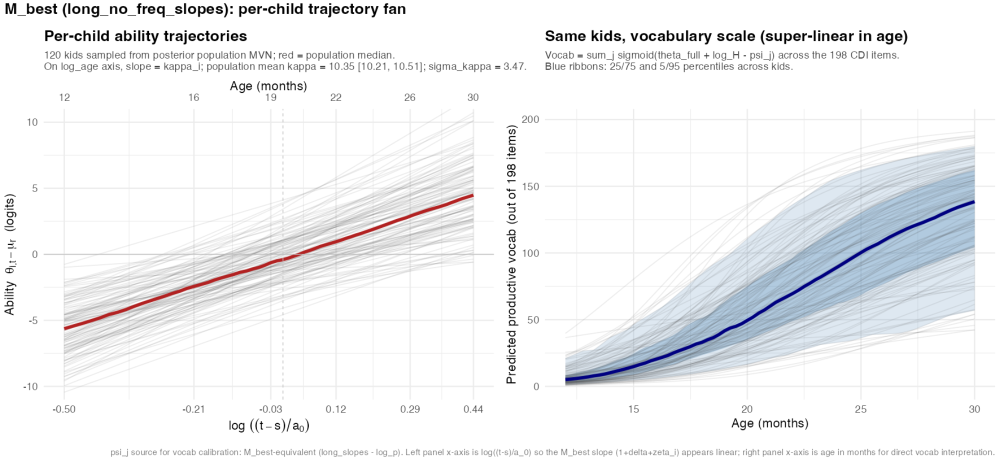
Each addition matters: 4-panel comparison to data

Variants fit to Wordbank English WS; predicted percentile fans vs. observed

quantile_demo.png

M_best fit: per-child trajectory fan

120 kids drawn from posterior of `long_no_freq_slopes`



Left: ability $\theta_{it} - \mu_r$ vs. $\log_{age}$, slope $= \kappa_i$, mean ≈ 10.4. **Right:** same kids, vocabulary on

M_best parameters

Variability and acceleration both bounded firmly above zero

Parameter	Posterior	What it does
σ_α	1.71 [1.55, 1.91]	Between-child efficiency variation.
$\boldsymbol{\pi_\alpha}$	**0.91 [0.89, 0.93]**	**Share of intercept variance from efficiency, not input.**
$\boldsymbol{\kappa}_{\text{pop}}$	**10.35 [10.21, 10.51]**	**Population scaling exponent: ability scales as (tokens)$^\kappa$.**
$\boldsymbol{\sigma_\kappa}$	**3.47 [3.14, 3.84]**	**Between-child variation in scaling exponent.**
$\rho(\xi, \kappa)$	+0.35 [+0.22, +0.47]	Higher-baseline kids have steeper growth.

Variability

$\pi_\alpha = 0.91$: efficiency variance dominates input variance.

Acceleration

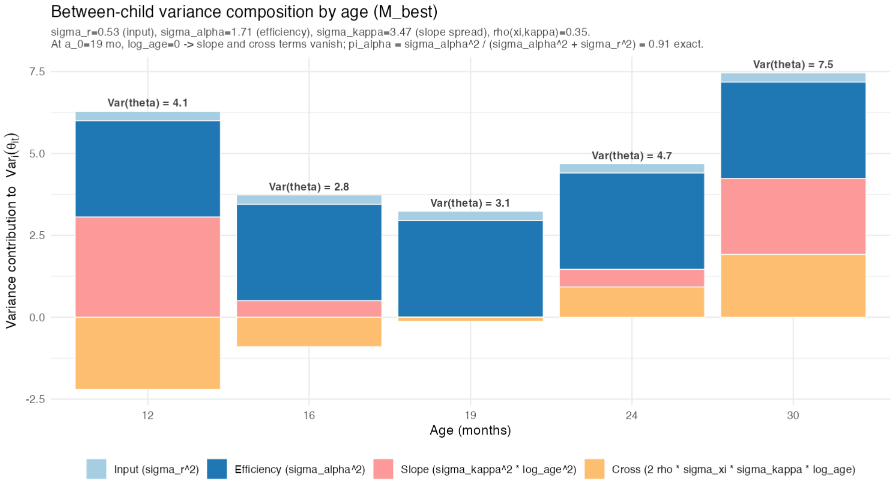
$\kappa_{\text{pop}} \approx 10.4 \gg 1$: scaling is wildly super-linear, far from the unit accumulator.

Variation in scaling

$\sigma_\kappa > \sigma_\alpha$: kids vary more in scaling exponent than in baseline.

Where between-child variance comes from, by age

π_α is exact at a_0 and qualified elsewhere



Cross-sample variability: π_α replicates

Five samples, two languages, three input-observation channels

Sample	N	σ_α	π_α
English Wordbank longitudinal	200	1.83 [1.65, 2.04]	0.91 [0.90, 0.93]
Norwegian Wordbank longitudinal	200	2.10 [1.90, 2.35]	0.94 [0.93, 0.95]
BabyView (head-mounted video)	20	1.13 [0.86, 1.61]	0.84 [0.68, 0.93]
SEEDLingS (LENA + comp correction)	44	1.39 [1.13, 1.74]	0.94 [0.89, 0.97]
Peekbank-Stanford (LWL processing)	62	1.64 [1.34, 2.03]	0.90 [0.86, 0.94]

Headline. $\pi_\alpha \in [0.84, 0.94]$ across every well-conditioned sample.
Input rate quantity explains $<10\%$ of between-child intercept variance.

Cross-channel replication: directly observed input (BabyView, SEEDLingS), inferred input (Wordbank), processing-coupled (Peekbank). SEEDLingS extreme corrected for parent-report noise via comprehension channel.

Cross-language acceleration: where it lives, not whether it does

Sample	admins/kid	$\boldsymbol{\kappa}_{\text{pop}}$	$\boldsymbol{\sigma_\kappa}$	kid range ($\kappa_{\textbf{pop}} \pm 2\sigma_\kappa$)
English	median 3	10.4 [10.2, 10.5]	3.5 [3.1, 3.8]	$\sim$3 to 17
Norwegian	median 8	12.4 [12.3, 12.6]	3.7 [3.4, 4.2]	$\sim$5 to 20

Population scaling exponent $\kappa_{\textbf{pop}} \sim 10-12$ in both languages. Implied $P(\kappa_i > 1) > 0.997$ in both.

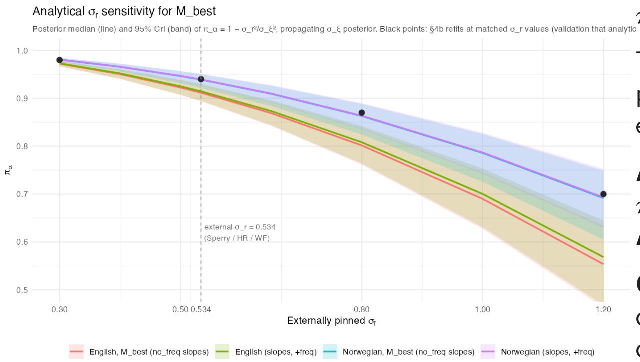
Internal decomposition into δ vs. ζ_i is identifiability-dependent on longitudinal density. In Norwegian (8 admins/kid) per-child slopes are well-identified, and the population term δ alone — without per-child deviations — *hurts* fit by 235 ELPD. In English (3 admins/kid), δ pools what Norwegian's data lets fall into ζ_i.

Reading. The acceleration signature is universal. The parameterization that captures it is identifiability-dependent on data density.

Concretely: in any sufficiently dense longitudinal sample, between-child κ_i spans roughly $5-15$.

Stress test 1: how robust is π_α to the input-rate prior?

Analytical sensitivity, validated by refits



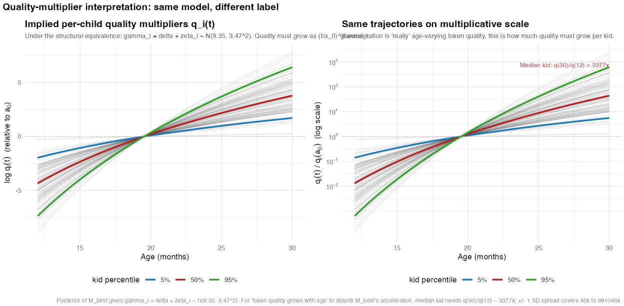
$\pi_\alpha(\sigma_r) = 1 - \sigma_r^2/\sigma_\xi^2$.

The data identifies σ_ξ^2 tightly; the σ_r prior just *partitions* it into input vs. efficiency.

At external $\sigma_r = 0.534$: $\pi_\alpha \approx 0.91 - 0.94$.
At doubled $\sigma_r = 1.2$: π_α still > 0.55.

Conclusion. Efficiency-dominated decomposition holds across any defensible external σ_r.

Stress test 2: can input quality absorb acceleration?



Reviewer reflex. "Maybe kids' tokens differ in *quality*, and that's what looks like acceleration."

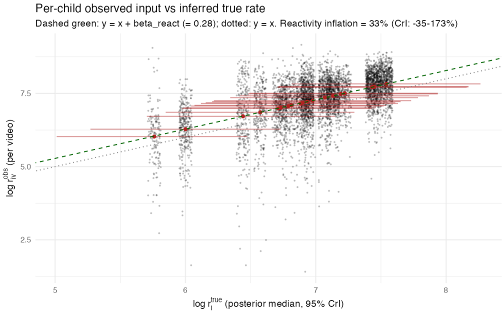
Structural answer. Per-child age-varying quality $q_i(t) = (t/a_0)^{\gamma_i}$ on input is observationally equivalent to $\gamma_i = \kappa_i - 1$.

Implication. Quality cannot escape acceleration — it *relabels* it. For the median kid, "quality" would have to grow $\sim$ 3,000$\times$ from age 12 to 30 mo. For the 95th percentile kid, $\sim 10^6\times$. Implausible as a substantive quality story.

Replicating with directly observed input

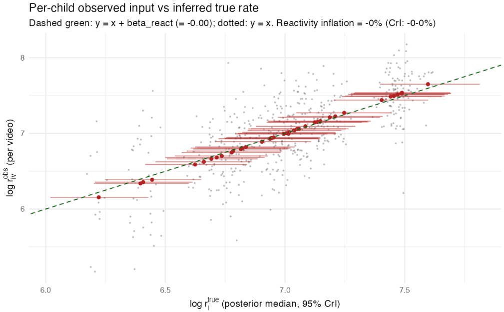
BabyView (head-mounted video) and SEEDLingS (LENA all-day audio)

BabyView, $N = 20$



$\pi_\alpha = 0.84\ [0.68, 0.93]$

SEEDLingS, $N = 44$ **(with comp correction)**



$\pi_\alpha = 0.94\ [0.89, 0.97]$

With per-recording observed input rates, the π_α headline holds. Input-rate variance is a small

Adding a processing-speed readout: Peekbank LWL channel

$N = 62$ **Stanford-linked subjects with both CDI and looking-while-listening**

Joint model adds a per-child LWL log-RT readout coupled to $\log \alpha_i$ via γ_{rt}.

Posterior.

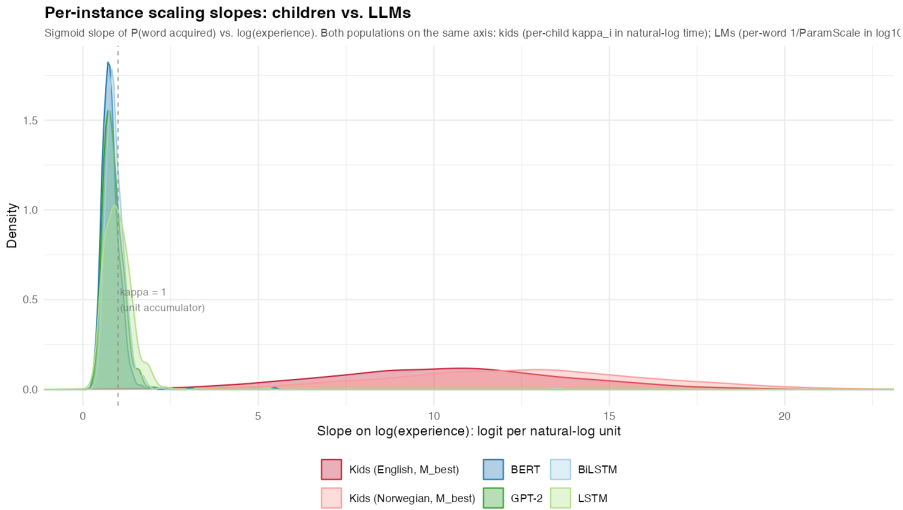
- $\gamma_{rt} = 0.082$ $[0.046, 0.125]$ — LWL processing speed is real, and bounded above zero.
- $\pi_\alpha = 0.90$ $[0.86, 0.94]$ — replicates within the Peekbank subset.
- $\rho(\kappa_i, \mathrm{rtslope}_i) = -0.02$ $[-0.33, +0.27]$ — CDI-side and LWL-side growth rates do *not* share variance.

Two clocks, not one. The acceleration in vocab growth and the maturation of processing speed are independent maturation processes within the same child — different "efficiency clocks" ticking in parallel.

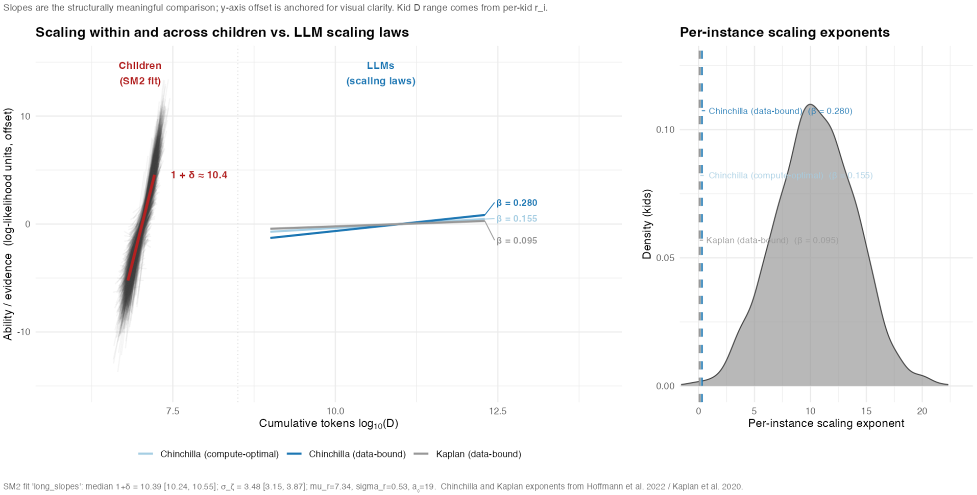
This rules out the simplest unification ("one underlying efficiency axis"); the two readouts each pick up a different developmental process.

Direct comparison: per-instance scaling slopes

Same task (CDI word acquisition), same axis (logit per log-experience)



Coda: scaling regimes are categorically different



Mechanisms not in the standard model

What human learning needs that pure accumulators don't have

Sources of variability

- Interactional input quality (joint attention, contingent talk)
- Genetic / heritable variation
- Environmental richness not captured by token quantity
- Processing-side individual differences (working memory, attention)

Sources of acceleration

- Growing processing efficiency with maturation
- Learning-to-learn (mutual exclusivity, shape bias, syntactic bootstrapping)
- Domain-general changes in attention, memory, perception
- Cross-word leverage / vocabulary-mediated bootstrapping

The structural finding constrains theory: any account of early word learning must produce both super-linear-in-tokens growth and large per-child variation in growth rate.

Summary

- Vocabulary uniquely affords *absolute* variability measurement — there is a true zero, and word counts are interpretable in word-count units.
- Two robust signatures across ∼30 languages and 5 modeling samples: **variability** ($\pi_\alpha \approx 0.91$, growing with age) and **acceleration** ($1 + \delta + \zeta_i \sim 10$, varying widely across kids).
- Both signatures violate a pure-accumulator null. Posited "quality" alternatives *relabel* acceleration rather than escape it.
- The picture is structurally different from LLM scaling laws on both axes (steeper exponent, between-instance variance).
- Open question: *which* mechanism — maturation, learning-to-learn, processing speed — is the load-bearing source of the acceleration we measure?

github.com/langcog/standard-model-2